

ASSIGNMENT 14.4

Name: K.Sanjana

Hall Ticket: 2303A52139

Batch: 41

Task 1: Responsive Homepage Layout for a Startup Website

Prompt: Create a clean, modern, and fully responsive startup website using HTML, CSS, and JavaScript that includes a header with navigation links such as Home, About, Services, and Contact, along with a hero section featuring a strong headline and a “Get Started” button. The website should also have a services section showcasing interactive feature cards for Product Strategy, UI/UX Design, Full-Stack Development, and Growth & Analytics, as well as a footer with contact information. Use JavaScript to enable navigation so that clicking on the navigation links redirects to their respective pages (about.html, services.html, contact.html), the “Get Started” button redirects appropriately, and each service card leads to its own page (product.html, design.html, development.html, analytics.html). The design should include hover or click-based color highlights for buttons and cards, maintain a professional and minimal appearance, and be fully responsive across desktop, tablet, and mobile devices using CSS media queries.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>InnovateX - Startup Solutions</title>

<link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700&display=swap"
rel="stylesheet">

<style>

:root{
--primary:#1d3557;
--accent:#e63946;
--bg:#f5f7fa;
--card:#ffffff;
}

*{
margin:0;
padding:0;
box-sizing:border-box;
}

body{
font-family:'Inter',sans-serif;
background:var(--bg);
color:#333;
}

/* HEADER */

header{
```

```
background:white;
box-shadow:0 3px 10px rgba(0,0,0,0.08);
padding:18px 40px;
}

.navbar{
display:flex;
justify-content:space-between;
align-items:center;
max-width:1200px;
margin:auto;
}

.logo{
font-size:28px;
font-weight:700;
color:var(--primary);
}

nav ul{
display:flex;
gap:30px;
list-style:none;
}

nav a{
text-decoration:none;
color:#333;
font-weight:600;
padding:6px 12px;
border-radius:6px;
transition:0.2s;
}

nav a:hover{
background:var(--primary);
color:white;
}

/* HERO */

.hero{
text-align:center;
padding:100px 20px;
background:linear-gradient(90deg,#e8efff,#f7f9fc);
}

.hero h1{
font-size:48px;
margin-bottom:20px;
color:var(--primary);
}

.hero p{
```

```
max-width:650px;
margin:auto;
margin-bottom:30px;
font-size:18px;
}

.cta-btn{
display:inline-block;
background:var(--primary);
color:white;
padding:14px 34px;
border-radius:8px;
text-decoration:none;
font-weight:600;
transition:0.2s;
}

.cta-btn:hover{
background:#14263e;
}

/* SERVICES */

main{
max-width:1200px;
margin:auto;
padding:60px 20px;
}

.features{
display:grid;
grid-template-columns:repeat(auto-fit,minmax(260px,1fr));
gap:30px;
}

.feature-card{
background:var(--card);
padding:35px;
border-radius:16px;
box-shadow:0 6px 25px rgba(0,0,0,0.08);
transition:0.25s;
cursor:pointer;
}

.feature-card:hover{
transform:translateY(-6px);
border:2px solid var(--accent);
}

.feature-title{
font-size:22px;
font-weight:700;
margin-bottom:12px;
color:var(--primary);
```

```
}

.feature-desc{
font-size:15px;
color:#555;
}

/* FOOTER */

footer{
background:white;
text-align:center;
padding:30px;
margin-top:60px;
box-shadow:0 -3px 10px rgba(0,0,0,0.05);
}

/* MOBILE */

@media(max-width:768px){

.hero h1{
font-size:32px;
}

.navbar{
flex-direction:column;
gap:10px;
}

}

</style>
</head>

<body>

<header>
<div class="navbar">

<div class="logo">InnovateX</div>

<nav>
<ul>
<li><a href="index.html">Home</a></li>
<li><a href="about.html">About</a></li>
<li><a href="services.html">Services</a></li>
<li><a href="contact.html">Contact</a></li>
</ul>
</nav>

</div>
</header>
```

```
<section class="hero">

<h1>Empowering Startups to Succeed</h1>

<p>
InnovateX delivers cutting-edge digital solutions to help startups launch, grow, and
scale.
From product strategy to seamless development, we turn your vision into reality.
</p>

<a href="services.html" class="cta-btn">Get Started</a>

</section>

<main>

<section class="features">

<div class="feature-card" onclick="location.href='product.html'">
<div class="feature-title">Product Strategy</div>
<div class="feature-desc">
We help define and validate your product vision with research and strategic planning.
</div>
</div>

<div class="feature-card" onclick="location.href='design.html'">
<div class="feature-title">UI / UX Design</div>
<div class="feature-desc">
Our designers create intuitive interfaces that improve user engagement.
</div>
</div>

<div class="feature-card" onclick="location.href='development.html'">
<div class="feature-title">Full-Stack Development</div>
<div class="feature-desc">
We build secure and scalable web and mobile applications.
</div>
</div>

<div class="feature-card" onclick="location.href='analytics.html'">
<div class="feature-title">Growth & Analytics</div>
<div class="feature-desc">
Data-driven marketing and analytics to maximize business growth.
</div>
</div>

</section>

</main>

<footer>
```

```

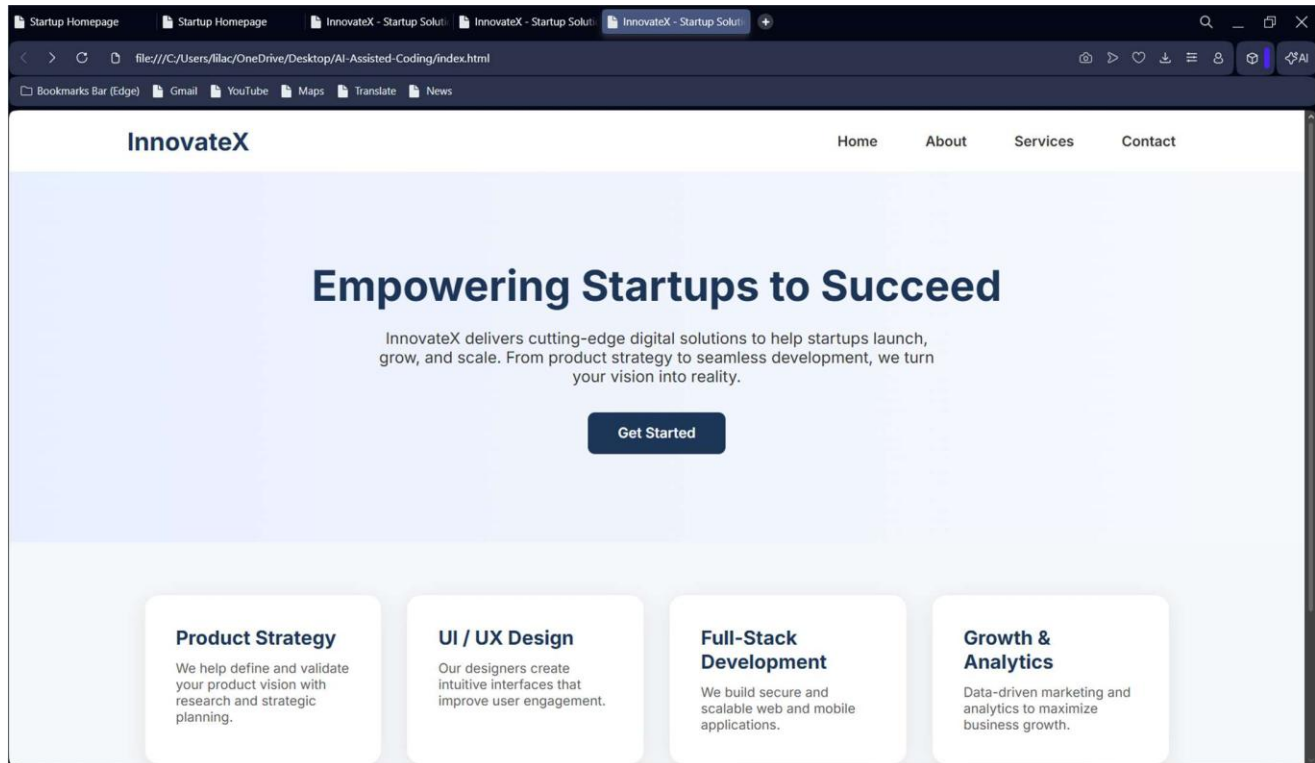
<p>© 2026 InnovateX. All rights reserved.</p>
<p>Contact: hello@innovatex.com | +1 (555) 123-4567</p>

</footer>

</body>
</html>

```

Output:



Link:

file:///C:/Users/lilac/OneDrive/Desktop/AI-Assisted-Coding/index.html

Explanation:

The code builds a responsive startup website using HTML for structure, CSS for styling, and JavaScript for interactivity. The header contains navigation links (Home, About, Services, Contact), the hero section introduces the startup with a Get Started button, and the services section displays clickable feature cards such as Product Strategy, UI/UX Design, Full-Stack Development, and Growth & Analytics.

Task 2: Interactive Call-to-Action Button

Prompt:

Create a modern responsive startup website using HTML, CSS, and JavaScript. The webpage should include a header with navigation links (Home, About, Services, Contact), a hero section with a headline, description, and a Get Started call-to-action button, a services section with clickable cards (Product Strategy, UI/UX Design, Full-Stack Development, Growth & Analytics), and a footer with contact information. Use JavaScript so that clicking navigation links, the Get Started button, or service cards redirects to separate pages (about.html, services.html, contact.html, product.html, design.html, development.html, analytics.html). Add hover and click color highlighting effects for buttons and cards, keep the design professional without emojis, and ensure the layout is responsive using CSS media queries for desktop, tablet, and mobile devices. Also include clear comments in the JavaScript explaining how the navigation and click interactions work, and keep the code clean and well-structured.

Code:

Style.css

```
:root{
  --bg: #0b1220;
  --surface: rgba(255,255,255,0.06);
  --surface-strong: rgba(255,255,255,0.10);
  --text: #eaf0ff;
  --muted: rgba(234,240,255,0.78);
  --border: rgba(234,240,255,0.14);
  --primary: #6ea8ff;
  --primary-strong: #3d7eff;
  --accent: #ff5470;
  --shadow: 0 18px 50px rgba(0,0,0,0.35);
  --radius: 16px;
}

*{ box-sizing: border-box; }

html, body{
  height: 100%;
}

body{
  margin: 0;
  font-family: ui-sans-serif, system-ui, -apple-system, "Segoe UI", Roboto, Arial,
"Inter", sans-serif;
  color: var(--text);
  background:
    radial-gradient(1100px 550px at 20% -10%, rgba(110,168,255,0.28), transparent 70%),
    radial-gradient(900px 520px at 90% 10%, rgba(255,84,112,0.18), transparent 65%),
    linear-gradient(180deg, #070b14 0%, var(--bg) 40%, #070b14 100%);
}

a{ color: inherit; text-decoration: none; }

.container{
  width: min(1120px, calc(100% - 48px));
  margin: 0 auto;
}

/* Header */
.site-header{
  position: sticky;
  top: 0;
  z-index: 50;
  backdrop-filter: blur(14px);
  background: rgba(7,11,20,0.6);
  border-bottom: 1px solid var(--border);
}

.nav{
  display: flex;
```

```
    align-items: center;
    justify-content: space-between;
    padding: 16px 0;
    gap: 16px;
}

.brand{
    display: flex;
    align-items: center;
    gap: 10px;
    font-weight: 800;
    letter-spacing: 0.2px;
}

.brand-mark{
    width: 34px;
    height: 34px;
    border-radius: 11px;
    background: linear-gradient(135deg, var(--primary) 0%, var(--accent) 100%);
    box-shadow: 0 10px 30px rgba(110,168,255,0.22);
}

.brand-name{
    font-size: 18px;
}

.nav-links{
    display: flex;
    align-items: center;
    gap: 8px;
    padding: 6px;
    border: 1px solid var(--border);
    border-radius: 999px;
    background: rgba(255,255,255,0.03);
}

.nav-link{
    display: inline-flex;
    align-items: center;
    justify-content: center;
    padding: 10px 14px;
    border-radius: 999px;
    color: var(--muted);
    font-weight: 650;
    transition: background 160ms ease, color 160ms ease, transform 160ms ease;
}

.nav-link:hover{
    background: rgba(255,255,255,0.06);
    color: var(--text);
}

.nav-link.is-active{
    background: rgba(110,168,255,0.18);
```



```
    color: var(--text);
    border: 1px solid rgba(110,168,255,0.35);
}

/* Hero */
.hero{
    padding: 86px 0 34px;
}

.hero-grid{
    display: grid;
    grid-template-columns: 1.2fr 0.8fr;
    gap: 28px;
    align-items: center;
}

.eyebrow{
    display: inline-flex;
    gap: 10px;
    align-items: center;
    padding: 8px 12px;
    border-radius: 999px;
    border: 1px solid var(--border);
    background: rgba(255,255,255,0.03);
    color: var(--muted);
    font-weight: 600;
    font-size: 13px;
}

.hero h1{
    margin: 16px 0 12px;
    font-size: clamp(34px, 4vw, 54px);
    line-height: 1.06;
    letter-spacing: -0.8px;
}

.hero p{
    margin: 0 0 20px;
    color: var(--muted);
    font-size: 16px;
    line-height: 1.6;
    max-width: 58ch;
}

.hero-actions{
    display: flex;
    gap: 12px;
    align-items: center;
    flex-wrap: wrap;
}

.btn{
    appearance: none;
    border: 1px solid transparent;
```

```

background: rgba(255,255,255,0.04);
color: var(--text);
padding: 12px 16px;
border-radius: 12px;
font-weight: 700;
cursor: pointer;
transition: transform 160ms ease, background 160ms ease, border-color 160ms ease, box-
shadow 160ms ease;
}

.btn:hover{
  transform: translateY(-1px);
  background: rgba(255,255,255,0.06);
  border-color: rgba(255,255,255,0.10);
}

.btn:active,
.btn.is-pressed{
  transform: translateY(0px) scale(0.99);
}

.btn-primary{
  background: linear-gradient(135deg, var(--primary-strong) 0%, var(--primary) 100%);
  border-color: rgba(110,168,255,0.35);
  box-shadow: 0 16px 50px rgba(61,126,255,0.24);
}

.btn-primary:hover{
  background: linear-gradient(135deg, #2f73ff 0%, #7ab2ff 100%);
}

.hero-panel{
  border: 1px solid var(--border);
  background: linear-gradient(180deg, rgba(255,255,255,0.06), rgba(255,255,255,0.03));
  border-radius: var(--radius);
  box-shadow: var(--shadow);
  padding: 18px;
}

.hero-panel .metric{
  display: grid;
  grid-template-columns: 1fr auto;
  gap: 6px;
  padding: 12px;
  border-radius: 14px;
  border: 1px solid rgba(255,255,255,0.10);
  background: rgba(255,255,255,0.03);
  margin-bottom: 10px;
}

.metric strong{ font-size: 14px; }
.metric span{ color: var(--muted); font-size: 13px; }

/* Section */

```

```
.section{
  padding: 34px 0 64px;
}

.section-header{
  display: flex;
  align-items: flex-end;
  justify-content: space-between;
  gap: 14px;
  margin-bottom: 18px;
}

.section-header h2{
  margin: 0;
  font-size: 22px;
  letter-spacing: -0.2px;
}

.section-header p{
  margin: 0;
  color: var(--muted);
  max-width: 60ch;
  font-size: 14px;
  line-height: 1.55;
}

.cards{
  display: grid;
  grid-template-columns: repeat(4, minmax(0, 1fr));
  gap: 16px;
}

.card{
  border: 1px solid var(--border);
  background: rgba(255,255,255,0.03);
  border-radius: var(--radius);
  padding: 18px;
  box-shadow: 0 10px 40px rgba(0,0,0,0.18);
  cursor: pointer;
  transition: transform 160ms ease, border-color 160ms ease, background 160ms ease;
  min-height: 150px;
  display: grid;
  gap: 10px;
  align-content: start;
}

.card:hover{
  transform: translateY(-3px);
  background: rgba(255,255,255,0.05);
  border-color: rgba(255,84,112,0.35);
}

.card:active,
.card.is-pressed{
```

```

    transform: translateY(0px) scale(0.99);
    border-color: rgba(110,168,255,0.4);
}

.card-title{
    font-weight: 800;
    letter-spacing: -0.2px;
}

.card-desc{
    color: var(--muted);
    font-size: 14px;
    line-height: 1.55;
}

/* Footer */
.site-footer{
    border-top: 1px solid var(--border);
    background: rgba(7,11,20,0.6);
    padding: 26px 0;
}

.footer-grid{
    display: grid;
    grid-template-columns: 1fr auto;
    gap: 18px;
    align-items: center;
}

.footer-small{
    color: var(--muted);
    font-size: 13px;
    line-height: 1.5;
}

.footer-contact{
    text-align: right;
    color: var(--muted);
    font-size: 13px;
}

/* Responsive */
@media (max-width: 980px){
    .hero-grid{ grid-template-columns: 1fr; }
    .cards{ grid-template-columns: repeat(2, minmax(0, 1fr)); }
    .footer-grid{ grid-template-columns: 1fr; }
    .footer-contact{ text-align: left; }
}

@media (max-width: 620px){
    .container{ width: min(1120px, calc(100% - 32px)); }
    .nav{ padding: 14px 0; }
    .nav-links{ width: 100%; justify-content: space-between; }
    .nav-link{ flex: 1; padding: 10px 10px; font-size: 14px; }

```

```
.cards{ grid-template-columns: 1fr; }
.hero{ padding-top: 64px; }
}
```

Script.js

```
/**
 * Shared navigation + click interactions for the multi-page site.
 *
 * How it works:
 * - We put `data-href="somepage.html"` on any element we want to behave like a link.
 * - This script attaches click handlers that:
 *   1) add a short "pressed" visual state (CSS class) for feedback
 *   2) redirect the browser to the page in `data-href`
 *
 * Why not inline onclick?
 * - Keeps HTML cleaner
 * - Centralizes behavior in one place
 * - Easier to maintain across multiple pages
 */

function setActiveNavLink() {
  const current = window.location.pathname.split("/").pop() || "index.html";
  const links = document.querySelectorAll("[data-nav]");

  links.forEach((link) => {
    const href = link.getAttribute("href") || "";
    const isActive = href.endsWith(current);
    link.classList.toggle("is-active", isActive);
  });
}

function addPressFeedback(el) {
  el.classList.add("is-pressed");
  window.setTimeout(() => el.classList.remove("is-pressed"), 140);
}

function wireClickableElements() {
  const clickables = document.querySelectorAll("[data-href]");

  clickables.forEach((el) => {
    el.addEventListener("click", () => {
      const target = el.getAttribute("data-href");
      if (!target) return;
      addPressFeedback(el);
      window.setTimeout(() => {
        window.location.href = target;
      }, 120);
    });

    // Optional keyboard accessibility: Enter/Space triggers click
    el.addEventListener("keydown", (evt) => {
      if (evt.key === "Enter" || evt.key === " ") {

```

```

        evt.preventDefault();
        el.click();
    }
});

// If it's not a native interactive element, make it keyboard focusable.
const tag = el.tagName.toLowerCase();
const isNaturallyFocusable = ["a", "button", "input"].includes(tag);
if (!isNaturallyFocusable && !el.hasAttribute("tabindex")) {
    el.setAttribute("tabindex", "0");
    el.setAttribute("role", "link");
}
});
}

document.addEventListener("DOMContentLoaded", () => {
    setActiveNavLink();
    wireClickableElements();
});

```

Index.html

```

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">

<title>InnovateX - Startup Solutions</title>

<link href="https://fonts.googleapis.com/css2?family=Inter:wght@400;600;700&display=swap"
rel="stylesheet">
<link rel="stylesheet" href="styles.css">
</head>

<body>

<header class="site-header">
  <div class="container">
    <div class="nav">
      <a class="brand" href="index.html" aria-label="InnovateX Home">
        <span class="brand-mark" aria-hidden="true"></span>
        <span class="brand-name">InnovateX</span>
      </a>

      <nav class="nav-links" aria-label="Primary navigation">
        <a class="nav-link" data-nav href="index.html">Home</a>
        <a class="nav-link" data-nav href="about.html">About</a>
        <a class="nav-link" data-nav href="services.html">Services</a>
        <a class="nav-link" data-nav href="contact.html">Contact</a>
      </nav>
    </div>
  </div>
</div>

```

```

</header>

<section class="hero">
  <div class="container">
    <div class="hero-grid">
      <div>
        <div class="eyebrow">Startup studio for product, design, and engineering</div>
        <h1>Build a product people trust—and a team that can ship it.</h1>
        <p>
          InnovateX helps founders turn ideas into market-ready products with clear
strategy, high-quality UI/UX,
          full-stack delivery, and measurable growth programs.
        </p>
        <div class="hero-actions">
          <button class="btn btn-primary" type="button" data-href="services.html">Get
Started</button>
          <button class="btn" type="button" data-href="about.html">How we work</button>
        </div>
      </div>

      <aside class="hero-panel" aria-label="Highlights">
        <div class="metric">
          <strong>Strategy → Launch</strong>
          <span>Roadmaps that reduce risk</span>
        </div>
        <div class="metric">
          <strong>Design System</strong>
          <span>Consistent UI at speed</span>
        </div>
        <div class="metric">
          <strong>Full-Stack Delivery</strong>
          <span>Secure, scalable builds</span>
        </div>
        <div class="metric" style="margin-bottom:0;">
          <strong>Growth Analytics</strong>
          <span>Decisions backed by data</span>
        </div>
      </aside>
    </div>
  </div>
</section>

<main class="section">
  <div class="container">
    <div class="section-header">
      <div>
        <h2>Services built for modern startups</h2>
        <p>Choose a track below. Every card is clickable and routes to a dedicated
page.</p>
      </div>
    </div>
  </div>

```

```

<section class="cards" aria-label="Services">
  <article class="card" data-href="product.html">
    <div class="card-title">Product Strategy</div>
    <div class="card-desc">Define the problem, validate assumptions, and plan a
roadmap your team can execute.</div>
  </article>

  <article class="card" data-href="design.html">
    <div class="card-title">UI/UX Design</div>
    <div class="card-desc">Design systems and experiences that feel premium and
convert users into customers.</div>
  </article>

  <article class="card" data-href="development.html">
    <div class="card-title">Full-Stack Development</div>
    <div class="card-desc">Build secure, scalable web apps with clean architecture and
fast iterations.</div>
  </article>

  <article class="card" data-href="analytics.html">
    <div class="card-title">Growth & Analytics</div>
    <div class="card-desc">Instrumentation, funnels, and experiments to grow with
confidence—not guesses.</div>
  </article>
</section>
</div>
</main>

<footer class="site-footer">
  <div class="container">
    <div class="footer-grid">
      <div class="footer-small">
        <div style="font-weight:800; margin-bottom:6px;">InnovateX</div>
        <div>Professional startup services for strategy, design, engineering, and
growth.</div>
      </div>
      <div class="footer-contact">
        <div>hello@innovatex.com</div>
        <div>+1 (555) 123-4567</div>
      </div>
    </div>
  </div>
</footer>

<script src="script.js"></script>

</body>
</html>

```

about.html

```

<!DOCTYPE html>
<html lang="en">

```



```

<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>About - InnovateX</title>
  <link rel="stylesheet" href="styles.css" />
</head>
<body>
  <header class="site-header">
    <div class="container">
      <div class="nav">
        <a class="brand" href="index.html" aria-label="InnovateX Home">
          <span class="brand-mark" aria-hidden="true"></span>
          <span class="brand-name">InnovateX</span>
        </a>
        <nav class="nav-links" aria-label="Primary navigation">
          <a class="nav-link" data-nav href="index.html">Home</a>
          <a class="nav-link" data-nav href="about.html">About</a>
          <a class="nav-link" data-nav href="services.html">Services</a>
          <a class="nav-link" data-nav href="contact.html">Contact</a>
        </nav>
      </div>
    </div>
  </header>

  <main class="section">
    <div class="container">
      <div class="section-header">
        <div>
          <h2>About InnovateX</h2>
          <p>We help startups move from ambiguity to traction with a disciplined, end-to-end delivery process.</p>
        </div>
      </div>

      <section class="cards" aria-label="About highlights">
        <article class="card" data-href="services.html">
          <div class="card-title">Clarity First</div>
          <div class="card-desc">We start with constraints, goals, and success metrics so every sprint has a purpose.</div>
        </article>
        <article class="card" data-href="design.html">
          <div class="card-title">Design with Intent</div>
          <div class="card-desc">We design for usability and outcomes, not just aesthetics.</div>
        </article>
        <article class="card" data-href="development.html">
          <div class="card-title">Engineering Quality</div>
          <div class="card-desc">Maintainable codebases, secure defaults, and performance from day one.</div>
        </article>
        <article class="card" data-href="contact.html">
          <div class="card-title">Collaborative Delivery</div>
          <div class="card-desc">Weekly demos, transparent planning, and clear communication across stakeholders.</div>
        </article>
      </section>
    </div>
  </main>

```

```

        </article>
    </section>
</div>
</main>

<footer class="site-footer">
    <div class="container">
        <div class="footer-grid">
            <div class="footer-small">
                <div style="font-weight:800; margin-bottom:6px;">InnovateX</div>
                <div>Strategy, design, engineering, and growth—built for teams that want to
ship.</div>
            </div>
            <div class="footer-contact">
                <div>hello@innovatex.com</div>
                <div>+1 (555) 123-4567</div>
            </div>
        </div>
    </div>
</footer>

    <script src="script.js"></script>
</body>
</html>

```

Services.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Services - InnovateX</title>
    <link rel="stylesheet" href="styles.css" />
</head>
<body>
    <header class="site-header">
        <div class="container">
            <div class="nav">
                <a class="brand" href="index.html" aria-label="InnovateX Home">
                    <span class="brand-mark" aria-hidden="true"></span>
                    <span class="brand-name">InnovateX</span>
                </a>
                <nav class="nav-links" aria-label="Primary navigation">
                    <a class="nav-link" data-nav href="index.html">Home</a>
                    <a class="nav-link" data-nav href="about.html">About</a>
                    <a class="nav-link" data-nav href="services.html">Services</a>
                    <a class="nav-link" data-nav href="contact.html">Contact</a>
                </nav>
            </div>
        </div>
    </header>

```

```

<main class="section">
  <div class="container">
    <div class="section-header">
      <div>
        <h2>Services</h2>
        <p>Explore each service area. Click any card to open the detailed page.</p>
      </div>
    </div>

    <section class="cards" aria-label="Service list">
      <article class="card" data-href="product.html">
        <div class="card-title">Product Strategy</div>
        <div class="card-desc">Discovery, positioning, roadmaps, and MVP planning that
reduces risk.</div>
      </article>
      <article class="card" data-href="design.html">
        <div class="card-title">UI/UX Design</div>
        <div class="card-desc">User journeys, prototyping, and design systems for
consistent experiences.</div>
      </article>
      <article class="card" data-href="development.html">
        <div class="card-title">Full-Stack Development</div>
        <div class="card-desc">Frontend, backend, APIs, and deployments with secure
foundations.</div>
      </article>
      <article class="card" data-href="analytics.html">
        <div class="card-title">Growth & Analytics</div>
        <div class="card-desc">Instrumentation, dashboards, experimentation, and
performance marketing support.</div>
      </article>
    </section>
  </div>
</main>

<footer class="site-footer">
  <div class="container">
    <div class="footer-grid">
      <div class="footer-small">
        <div style="font-weight:800; margin-bottom:6px;">InnovateX</div>
        <div>Need a custom engagement? We can tailor a plan to your stage and
goals.</div>
      </div>
      <div class="footer-contact">
        <div>hello@innovatex.com</div>
        <div>+1 (555) 123-4567</div>
      </div>
    </div>
  </div>
</footer>

  <script src="script.js"></script>
</body>
</html>

```

Contact.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Contact - InnovateX</title>
  <link rel="stylesheet" href="styles.css" />
</head>
<body>
  <header class="site-header">
    <div class="container">
      <div class="nav">
        <a class="brand" href="index.html" aria-label="InnovateX Home">
          <span class="brand-mark" aria-hidden="true"></span>
          <span class="brand-name">InnovateX</span>
        </a>
        <nav class="nav-links" aria-label="Primary navigation">
          <a class="nav-link" data-nav href="index.html">Home</a>
          <a class="nav-link" data-nav href="about.html">About</a>
          <a class="nav-link" data-nav href="services.html">Services</a>
          <a class="nav-link" data-nav href="contact.html">Contact</a>
        </nav>
      </div>
    </div>
  </header>

  <main class="section">
    <div class="container">
      <div class="section-header">
        <div>
          <h2>Contact</h2>
          <p>Tell us what you're building. We'll reply with next steps and a simple
plan.</p>
        </div>
      </div>

      <section class="cards" aria-label="Contact options">
        <article class="card" data-href="services.html">
          <div class="card-title">Project Inquiry</div>
          <div class="card-desc">Start with scope, timeline, and desired outcomes. We'll
propose an engagement.</div>
        </article>
        <article class="card" data-href="about.html">
          <div class="card-title">How We Work</div>
          <div class="card-desc">Learn about our process, communication cadence, and
delivery standards.</div>
        </article>
        <article class="card" data-href="analytics.html">
          <div class="card-title">Analytics Audit</div>
          <div class="card-desc">Instrument events, validate funnels, and build dashboards
that your team trusts.</div>
        </article>
      </section>
    </div>
  </main>
</body>
</html>
```

```

        <article class="card" data-href="development.html">
          <div class="card-title">Engineering Consultation</div>
          <div class="card-desc">Architecture review, performance improvements, and
production readiness checks.</div>
        </article>
      </section>
    </div>
  </main>

  <footer class="site-footer">
    <div class="container">
      <div class="footer-grid">
        <div class="footer-small">
          <div style="font-weight:800; margin-bottom:6px;">InnovateX</div>
          <div>Preferred contact: email. Typical response time: 1 business day.</div>
        </div>
        <div class="footer-contact">
          <div>hello@innovatex.com</div>
          <div>+1 (555) 123-4567</div>
        </div>
      </div>
    </div>
  </footer>

  <script src="script.js"></script>
</body>
</html>

```

Product.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Product Strategy - InnovateX</title>
  <link rel="stylesheet" href="styles.css" />
</head>
<body>
  <header class="site-header">
    <div class="container">
      <div class="nav">
        <a class="brand" href="index.html" aria-label="InnovateX Home">
          <span class="brand-mark" aria-hidden="true"></span>
          <span class="brand-name">InnovateX</span>
        </a>
        <nav class="nav-links" aria-label="Primary navigation">
          <a class="nav-link" data-nav href="index.html">Home</a>
          <a class="nav-link" data-nav href="about.html">About</a>
          <a class="nav-link" data-nav href="services.html">Services</a>
          <a class="nav-link" data-nav href="contact.html">Contact</a>
        </nav>
      </div>
    </div>
  </header>

```

```

    </div>
</header>

<main class="section">
  <div class="container">
    <div class="section-header">
      <div>
        <h2>Product Strategy</h2>
        <p>Turn ideas into a roadmap with clear priorities, constraints, and measurable
outcomes.</p>
      </div>
    </div>

    <section class="cards" aria-label="Product strategy deliverables">
      <article class="card" data-href="contact.html">
        <div class="card-title">Discovery & Research</div>
        <div class="card-desc">Stakeholder interviews, user insights, and competitive
review to de-risk decisions.</div>
      </article>
      <article class="card" data-href="services.html">
        <div class="card-title">MVP Definition</div>
        <div class="card-desc">Scope the smallest valuable release and align it to a
realistic timeline.</div>
      </article>
      <article class="card" data-href="design.html">
        <div class="card-title">UX Flows</div>
        <div class="card-desc">Map the critical paths so design and engineering move
with shared context.</div>
      </article>
      <article class="card" data-href="development.html">
        <div class="card-title">Technical Plan</div>
        <div class="card-desc">Architecture notes, API planning, and delivery sequencing
for predictable execution.</div>
      </article>
    </section>
  </div>
</main>

<footer class="site-footer">
  <div class="container">
    <div class="footer-grid">
      <div class="footer-small">
        <div style="font-weight:800; margin-bottom:6px;">InnovateX</div>
        <div>Strategy engagements typically start with a 1-2 week discovery
sprint.</div>
      </div>
      <div class="footer-contact">
        <div>hello@innovatex.com</div>
        <div>+1 (555) 123-4567</div>
      </div>
    </div>
  </div>
</footer>

```

```
<script src="script.js"></script>
</body>
</html>
```

Design.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>UI/UX Design - InnovateX</title>
  <link rel="stylesheet" href="styles.css" />
</head>
<body>
  <header class="site-header">
    <div class="container">
      <div class="nav">
        <a class="brand" href="index.html" aria-label="InnovateX Home">
          <span class="brand-mark" aria-hidden="true"></span>
          <span class="brand-name">InnovateX</span>
        </a>
        <nav class="nav-links" aria-label="Primary navigation">
          <a class="nav-link" data-nav href="index.html">Home</a>
          <a class="nav-link" data-nav href="about.html">About</a>
          <a class="nav-link" data-nav href="services.html">Services</a>
          <a class="nav-link" data-nav href="contact.html">Contact</a>
        </nav>
      </div>
    </div>
  </header>

  <main class="section">
    <div class="container">
      <div class="section-header">
        <div>
          <h2>UI/UX Design</h2>
          <p>Design systems and product experiences that feel premium and convert.</p>
        </div>
      </div>

      <section class="cards" aria-label="Design services">
        <article class="card" data-href="product.html">
          <div class="card-title">User Journeys</div>
          <div class="card-desc">Define critical flows so every screen supports the user's
goal.</div>
        </article>
        <article class="card" data-href="services.html">
          <div class="card-title">Prototyping</div>
          <div class="card-desc">Clickable prototypes to validate direction before
engineering investment.</div>
        </article>
        <article class="card" data-href="development.html">
```

```

        <div class="card-title">Design System</div>
        <div class="card-desc">Reusable components and tokens for consistent UI across
platforms.</div>
    </article>
    <article class="card" data-href="contact.html">
        <div class="card-title">Usability Review</div>
        <div class="card-desc">Heuristic evaluation and actionable improvements for
clarity and conversion.</div>
    </article>
</section>
</div>
</main>

<footer class="site-footer">
    <div class="container">
        <div class="footer-grid">
            <div class="footer-small">
                <div style="font-weight:800; margin-bottom:6px;">InnovateX</div>
                <div>Design work is delivered in weekly milestones with review-ready
artifacts.</div>
            </div>
            <div class="footer-contact">
                <div>hello@innovatex.com</div>
                <div>+1 (555) 123-4567</div>
            </div>
        </div>
    </div>
</footer>

    <script src="script.js"></script>
</body>
</html>

```

Development.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Full-Stack Development - InnovateX</title>
    <link rel="stylesheet" href="styles.css" />
</head>
<body>
    <header class="site-header">
        <div class="container">
            <div class="nav">
                <a class="brand" href="index.html" aria-label="InnovateX Home">
                    <span class="brand-mark" aria-hidden="true"></span>
                    <span class="brand-name">InnovateX</span>
                </a>
                <nav class="nav-links" aria-label="Primary navigation">
                    <a class="nav-link" data-nav href="index.html">Home</a>

```



```

    <a class="nav-link" data-nav href="about.html">About</a>
    <a class="nav-link" data-nav href="services.html">Services</a>
    <a class="nav-link" data-nav href="contact.html">Contact</a>
  </nav>
</div>
</div>
</header>

<main class="section">
  <div class="container">
    <div class="section-header">
      <div>
        <h2>Full-Stack Development</h2>
        <p>Secure, scalable engineering that ships fast and remains maintainable.</p>
      </div>
    </div>

    <section class="cards" aria-label="Development capabilities">
      <article class="card" data-href="design.html">
        <div class="card-title">Frontend Engineering</div>
        <div class="card-desc">Accessible UI, performant rendering, and a component-
driven architecture.</div>
      </article>
      <article class="card" data-href="product.html">
        <div class="card-title">Backend & APIs</div>
        <div class="card-desc">Clean interfaces, auth, rate limits, and operational
resilience.</div>
      </article>
      <article class="card" data-href="analytics.html">
        <div class="card-title">Observability</div>
        <div class="card-desc">Logging, metrics, and tracing so issues are detected
early.</div>
      </article>
      <article class="card" data-href="contact.html">
        <div class="card-title">Deployment & Handoff</div>
        <div class="card-desc">CI/CD setup, environment configuration, and documentation
for your team.</div>
      </article>
    </section>
  </div>
</main>

<footer class="site-footer">
  <div class="container">
    <div class="footer-grid">
      <div class="footer-small">
        <div style="font-weight:800; margin-bottom:6px;">InnovateX</div>
        <div>We build with secure defaults and performance budgets from the start.</div>
      </div>
      <div class="footer-contact">
        <div>hello@innovatex.com</div>
        <div>+1 (555) 123-4567</div>
      </div>
    </div>
  </div>

```

```

    </div>
</footer>

<script src="script.js"></script>
</body>
</html>

```

Analytics.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <title>Growth & Analytics - InnovateX</title>
  <link rel="stylesheet" href="styles.css" />
</head>
<body>
  <header class="site-header">
    <div class="container">
      <div class="nav">
        <a class="brand" href="index.html" aria-label="InnovateX Home">
          <span class="brand-mark" aria-hidden="true"></span>
          <span class="brand-name">InnovateX</span>
        </a>
        <nav class="nav-links" aria-label="Primary navigation">
          <a class="nav-link" data-nav href="index.html">Home</a>
          <a class="nav-link" data-nav href="about.html">About</a>
          <a class="nav-link" data-nav href="services.html">Services</a>
          <a class="nav-link" data-nav href="contact.html">Contact</a>
        </nav>
      </div>
    </div>
  </header>

  <main class="section">
    <div class="container">
      <div class="section-header">
        <div>
          <h2>Growth & Analytics</h2>
          <p>Measure what matters, improve conversion, and make decisions with
confidence.</p>
        </div>
      </div>

      <section class="cards" aria-label="Analytics services">
        <article class="card" data-href="development.html">
          <div class="card-title">Event Instrumentation</div>
          <div class="card-desc">Define events, naming conventions, and tracking plans
that scale.</div>
        </article>
        <article class="card" data-href="services.html">
          <div class="card-title">Dashboards</div>

```

```

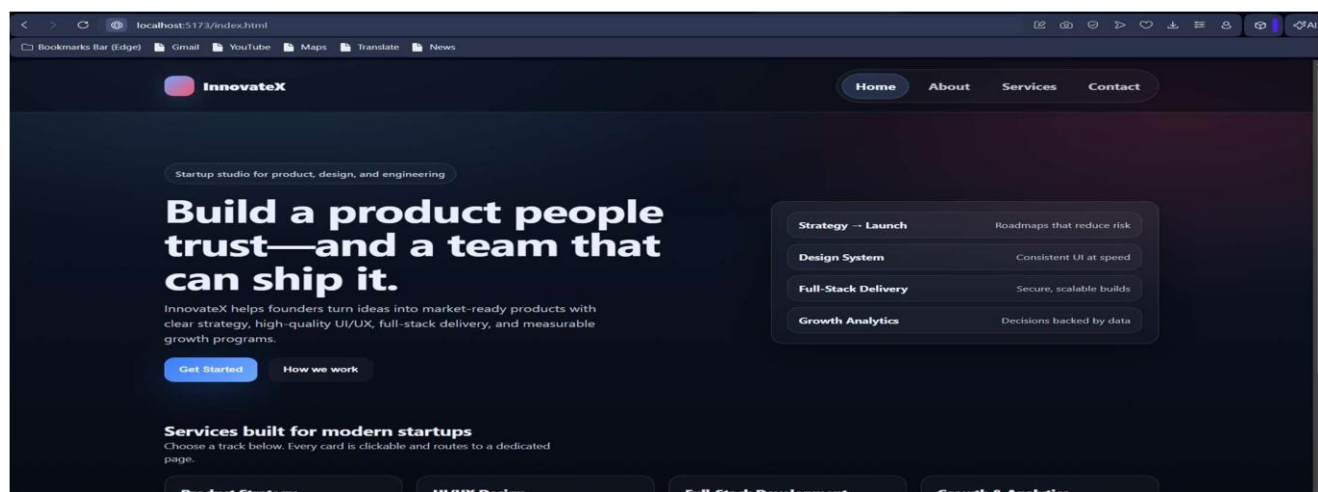
        <div class="card-desc">Clear KPIs, funnel views, and weekly reporting for
stakeholders.</div>
    </article>
    <article class="card" data-href="product.html">
        <div class="card-title">Experimentation</div>
        <div class="card-desc">A/B tests, rollout plans, and analysis to iterate
safely.</div>
    </article>
    <article class="card" data-href="contact.html">
        <div class="card-title">Growth Roadmap</div>
        <div class="card-desc">Channel strategy, content planning, and measurement
frameworks.</div>
    </article>
</section>
</div>
</main>

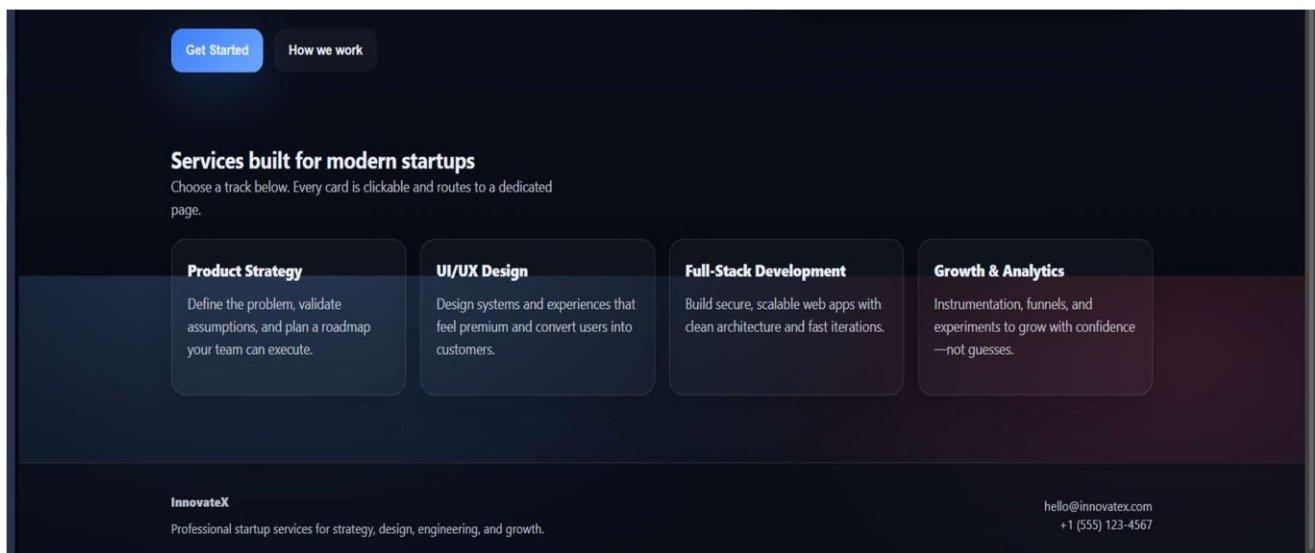
<footer class="site-footer">
    <div class="container">
        <div class="footer-grid">
            <div class="footer-small">
                <div style="font-weight:800; margin-bottom:6px;">InnovateX</div>
                <div>Analytics should be trustworthy, fast to interpret, and easy to
maintain.</div>
            </div>
            <div class="footer-contact">
                <div>hello@innovatex.com</div>
                <div>+1 (555) 123-4567</div>
            </div>
        </div>
    </div>
</footer>

<script src="script.js"></script>
</body>
</html>

```

Output:





Link:

<http://localhost:5173/index.html>

Explanation:

The code creates a responsive startup website using HTML for structure, CSS for styling, and JavaScript for interactivity. Navigation links, the Get Started button, and service cards use JavaScript or links to redirect users to different pages when clicked. CSS media queries make the layout responsive so the website works properly on desktop, tablet, and mobile devices.

Task 3: Contact Form with Client-Side Validation

Prompt:

Create a contact form webpage using HTML, CSS, and JavaScript for a company website. The form should include input fields for Name, Email, and Message along with a Submit button. Use JavaScript to perform client-side validation so that the form prevents empty submissions and checks whether the email entered follows a valid email format. If any field is invalid, display clear inline error messages near the respective input field, and prevent the form from submitting. If all inputs are valid, display a success message to the user without reloading the page. Ensure the design is clean, responsive, and the JavaScript code includes meaningful comments explaining the validation logic.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <title>Contact Us - Acme Corp</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <style>
    :root {
      --bg: #0f172a;
      --card: #020617;
      --accent: #3b82f6;
      --accent-soft: rgba(59,130,246,0.15);
      --error: #f97373;
      --text: #e5e7eb;
      --muted: #9ca3af;
      --border: #1f2937;
      --radius: 12px;
```

```

    --shadow: 0 18px 40px rgba(15,23,42,0.7);
  }

  * {
    box-sizing: border-box;
  }

  body {
    margin: 0;
    font-family: system-ui, -apple-system, BlinkMacSystemFont, "Segoe UI", sans-serif;
    background:
      radial-gradient(circle at top left, rgba(56,189,248,0.18), transparent 60%),
      radial-gradient(circle at bottom right, rgba(59,130,246,0.16), transparent 55%),
      var(--bg);
    color: var(--text);
    min-height: 100vh;
    display: flex;
    align-items: center;
    justify-content: center;
    padding: 24px 16px;
  }

  .page {
    width: 100%;
    max-width: 960px;
    display: grid;
    grid-template-columns: 1.1fr 0.9fr;
    gap: 32px;
    background: linear-gradient(135deg, rgba(15,23,42,0.96), rgba(15,23,42,0.98));
    border-radius: 24px;
    box-shadow: var(--shadow);
    border: 1px solid rgba(148,163,184,0.25);
    padding: 32px 32px 32px 32px;
  }

  .intro {
    padding-right: 8px;
    display: flex;
    flex-direction: column;
    justify-content: center;
    gap: 16px;
  }

  .badge {
    display: inline-flex;
    align-items: center;
    gap: 10px;
    padding: 6px 10px;
    border-radius: 999px;
    border: 1px solid rgba(148,163,184,0.4);
    background: rgba(15,23,42,0.8);
    color: var(--muted);
    font-size: 13px;
    font-weight: 500;
  }

```

```
}

.badge-dot {
  width: 8px;
  height: 8px;
  border-radius: 999px;
  background: #22c55e;
  box-shadow: 0 0 0 4px rgba(34,197,94,0.25);
}

h1 {
  margin: 0;
  font-size: clamp(26px, 3vw, 34px);
  letter-spacing: -0.03em;
}

.intro p {
  margin: 0;
  color: var(--muted);
  font-size: 14px;
  line-height: 1.6;
  max-width: 40rem;
}

.meta {
  display: flex;
  flex-wrap: wrap;
  gap: 16px;
  margin-top: 12px;
  font-size: 13px;
  color: var(--muted);
}

.meta span {
  padding: 6px 10px;
  border-radius: 999px;
  border: 1px solid rgba(148,163,184,0.28);
}

.card {
  background: radial-gradient(circle at top left, rgba(59,130,246,0.18), transparent
55%), var(--card);
  border-radius: var(--radius);
  padding: 22px 20px 20px;
  border: 1px solid rgba(148,163,184,0.35);
  display: flex;
  flex-direction: column;
  gap: 14px;
}

.card h2 {
  margin: 0 0 2px;
  font-size: 18px;
  letter-spacing: -0.01em;
}
```

```

}

.card > p {
  margin: 0;
  color: var(--muted);
  font-size: 13px;
}

form {
  display: flex;
  flex-direction: column;
  gap: 12px;
  margin-top: 4px;
}

label {
  display: block;
  font-size: 13px;
  margin-bottom: 4px;
  color: #e5e7eb;
}

.required {
  color: #f97373;
  margin-left: 2px;
}

input[type="text"],
input[type="email"],
textarea {
  width: 100%;
  padding: 9px 10px;
  border-radius: 9px;
  border: 1px solid var(--border);
  background: rgba(15,23,42,0.9);
  color: var(--text);
  font-size: 14px;
  outline: none;
  transition: border-color 150ms ease, box-shadow 150ms ease, background 150ms ease;
}

input::placeholder,
textarea::placeholder {
  color: #6b7280;
}

input:focus,
textarea:focus {
  border-color: rgba(59,130,246,0.7);
  box-shadow: 0 0 0 1px rgba(59,130,246,0.6);
  background: rgba(15,23,42,0.98);
}

.field-error input,

```

```
.field-error textarea {
  border-color: var(--error);
  box-shadow: 0 0 0 1px rgba(248,113,113,0.65);
}

.error-message {
  font-size: 12px;
  color: var(--error);
  margin-top: 3px;
  min-height: 14px;
}

.helper-text {
  font-size: 11px;
  color: var(--muted);
  margin-top: 3px;
}

textarea {
  resize: vertical;
  min-height: 90px;
  max-height: 220px;
}

.actions {
  display: flex;
  align-items: center;
  justify-content: space-between;
  gap: 12px;
  margin-top: 8px;
}

button[type="submit"] {
  padding: 9px 18px;
  border-radius: 999px;
  border: 1px solid rgba(37,99,235,0.9);
  background: radial-gradient(circle at top left, rgba(96,165,250,0.35), transparent
60%), var(--accent);
  color: white;
  font-weight: 600;
  font-size: 14px;
  cursor: pointer;
  display: inline-flex;
  align-items: center;
  gap: 6px;
  box-shadow: 0 12px 35px rgba(37,99,235,0.5);
  transition: transform 120ms ease, box-shadow 120ms ease, filter 120ms ease;
}

button[type="submit"]:hover {
  transform: translateY(-1px);
  filter: brightness(1.03);
}
```



```

button[type="submit"]:active {
  transform: translateY(0px) scale(0.99);
  box-shadow: 0 8px 24px rgba(37,99,235,0.4);
}

.status {
  font-size: 12px;
  color: var(--muted);
  min-height: 16px;
}

.status.success {
  color: #4ade80;
}

.status.error {
  color: var(--error);
}

@media (max-width: 800px) {
  .page {
    grid-template-columns: 1fr;
    padding: 24px 22px 26px;
  }
  .intro {
    padding-right: 0;
  }
}

@media (max-width: 480px) {
  body {
    padding: 16px 10px;
  }
  .page {
    padding: 20px 16px 20px;
    border-radius: 18px;
  }
  .actions {
    flex-direction: column;
    align-items: stretch;
  }
  button[type="submit"] {
    justify-content: center;
    width: 100%;
  }
  .status {
    text-align: center;
  }
}
</style>
</head>
<body>
  <div class="page">
    <!-- Left: intro / copy -->

```

```

<section class="intro">
  <div class="badge">
    <span class="badge-dot"></span>
    <span>Contact our team</span>
  </div>
  <h1>Let's talk about your next project.</h1>
  <p>
    Share a bit about your product, timelines, and what success looks like. We'll
review your message
    and respond with clear next steps—no spam, no pressure.
  </p>
  <div class="meta">
    <span>Response in 1-2 business days</span>
    <span>Strategy, design, and engineering inquiries</span>
  </div>
</section>

<!-- Right: form card -->
<section class="card" aria-label="Contact form">
  <h2>Send us a message</h2>
  <p>All fields are required. We'll keep your information confidential.</p>

  <form id="contactForm" novalidate>
    <!-- Name field -->
    <div class="field" id="field-name">
      <label for="name">Name<span class="required">*</span></label>
      <input type="text" id="name" name="name" placeholder="Jane Doe" />
      <div class="error-message" id="nameError"></div>
    </div>

    <!-- Email field -->
    <div class="field" id="field-email">
      <label for="email">Email<span class="required">*</span></label>
      <input type="email" id="email" name="email" placeholder="you@example.com" />
      <div class="error-message" id="emailError"></div>
      <div class="helper-text">We'll only use this to respond to your message.</div>
    </div>

    <!-- Message field -->
    <div class="field" id="field-message">
      <label for="message">Message<span class="required">*</span></label>
      <textarea id="message" name="message" placeholder="Tell us a bit about what
you're working on..."></textarea>
      <div class="error-message" id="messageError"></div>
    </div>

    <div class="actions">
      <button type="submit">
        Submit
      </button>
      <div class="status" id="formStatus" aria-live="polite"></div>
    </div>
  </form>
</section>

```

```

</div>

<script>
  // Simple email validation pattern:
  // - Some characters before '@'
  // - Then a domain with at least one dot
  // - Rough, but good enough for client-side checks
  function isValidEmail(email) {
    const pattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    return pattern.test(email);
  }

  // Utility to set an error message and add error styling to a field
  function setFieldError(fieldId, message) {
    const fieldWrapper = document.getElementById("field-" + fieldId);
    const errorEl = document.getElementById(fieldId + "Error");
    if (!fieldWrapper || !errorEl) return;

    if (message) {
      fieldWrapper.classList.add("field-error");
      errorEl.textContent = message;
    } else {
      fieldWrapper.classList.remove("field-error");
      errorEl.textContent = "";
    }
  }

  document.addEventListener("DOMContentLoaded", function () {
    const form = document.getElementById("contactForm");
    const nameInput = document.getElementById("name");
    const emailInput = document.getElementById("email");
    const messageInput = document.getElementById("message");
    const statusEl = document.getElementById("formStatus");

    form.addEventListener("submit", function (event) {
      event.preventDefault(); // Prevent form from actually submitting / reloading

      // Clear overall status and previous field errors
      statusEl.textContent = "";
      statusEl.className = "status";
      setFieldError("name", "");
      setFieldError("email", "");
      setFieldError("message", "");

      let hasError = false;

      // 1. Validate Name: must not be empty / whitespace only
      const nameValue = nameInput.value.trim();
      if (nameValue === "") {
        setFieldError("name", "Please enter your name.");
        hasError = true;
      }

      // 2. Validate Email: non-empty and matches pattern

```

```
const emailValue = emailInput.value.trim();
if (emailValue === "") {
    setFieldError("email", "Please enter your email address.");
    hasError = true;
} else if (!isValidEmail(emailValue)) {
    setFieldError("email", "Please enter a valid email address (e.g.,
you@example.com).");
    hasError = true;
}

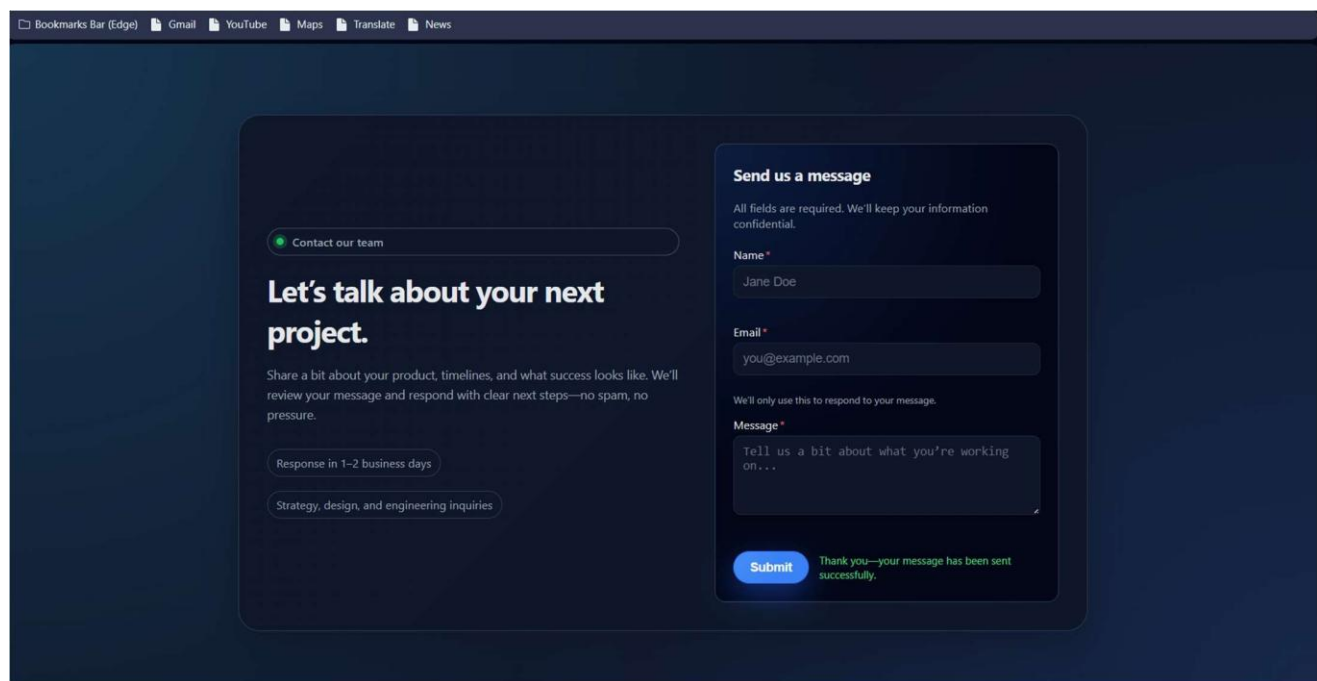
// 3. Validate Message: must not be empty / whitespace only
const messageValue = messageInput.value.trim();
if (messageValue === "") {
    setFieldError("message", "Please enter a message.");
    hasError = true;
}

// If any field is invalid, show a generic status and exit
if (hasError) {
    statusEl.textContent = "Please fix the errors above and try again.";
    statusEl.classList.add("error");
    return; // Do not proceed to "submit"
}

// If everything is valid:
// - Show success message
// - Optionally clear the inputs (simulate a successful send)
statusEl.textContent = "Thank you—your message has been sent successfully.";
statusEl.classList.add("success");

// Optional: clear the form fields after success
nameInput.value = "";
emailInput.value = "";
messageInput.value = "";
});
});
</script>
</body>
</html>
```

Output:



Link:

file:///C:/Users/lilac/OneDrive/Desktop/index3.html

Explanation:

The code creates a contact form using HTML for the structure and CSS for styling. JavaScript performs client-side validation to check if fields are empty and if the email format is valid. Error messages are shown for invalid input, and a success message appears when the form is correctly filled.

Task 4: Dynamic List Management for a Product Page

Prompt:

Create a simple product list webpage using HTML, CSS, and JavaScript. The page should display a list of items using an HTML list element and include two buttons labeled Add Item and Remove Item. Use JavaScript to dynamically update the list so that clicking the Add Item button adds a new item to the list and clicking the Remove Item button removes the last item from the list. The changes should update the DOM instantly without reloading the page. Ensure the design is clean and responsive, and include clear comments in the JavaScript explaining how the add and remove functionality works.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <title>Dynamic Product List</title>
  <meta name="viewport" content="width=device-width, initial-scale=1.0" />
  <style>
    :root {
      --bg: #0f172a;
      --card: #020617;
      --border: #1f2937;
      --accent: #3b82f6;
      --accent-soft: rgba(59,130,246,0.15);
```

```

--text: #e5e7eb;
--muted: #9ca3af;
--radius: 14px;
}

* {
  box-sizing: border-box;
}

body {
  margin: 0;
  min-height: 100vh;
  display: flex;
  align-items: center;
  justify-content: center;
  padding: 24px 16px;
  font-family: system-ui, -apple-system, BlinkMacSystemFont, "Segoe UI", sans-serif;
  color: var(--text);
  background:
    radial-gradient(circle at top left, rgba(96,165,250,0.2), transparent 55%),
    radial-gradient(circle at bottom right, rgba(14,165,233,0.18), transparent 55%),
    #020617;
}

.wrapper {
  width: 100%;
  max-width: 520px;
  background: linear-gradient(145deg, rgba(15,23,42,0.98), rgba(15,23,42,0.95));
  border-radius: 22px;
  padding: 22px 20px 20px;
  border: 1px solid rgba(148,163,184,0.35);
  box-shadow: 0 18px 40px rgba(15,23,42,0.85);
  display: flex;
  flex-direction: column;
  gap: 16px;
}

h1 {
  margin: 0;
  font-size: 22px;
  letter-spacing: -0.03em;
}

p {
  margin: 4px 0 0;
  font-size: 13px;
  color: var(--muted);
}

.controls {
  display: flex;
  gap: 10px;
  flex-wrap: wrap;
  margin-top: 10px;
}

```

```

}

button {
  border-radius: 999px;
  border: 1px solid rgba(59,130,246,0.85);
  background: radial-gradient(circle at top left, rgba(59,130,246,0.25), transparent
55%), var(--accent);
  color: white;
  font-size: 14px;
  font-weight: 600;
  padding: 8px 14px;
  cursor: pointer;
  display: inline-flex;
  align-items: center;
  gap: 6px;
  transition: transform 120ms ease, box-shadow 120ms ease, filter 120ms ease;
  box-shadow: 0 12px 30px rgba(37,99,235,0.5);
}

button.secondary {
  background: transparent;
  border-color: rgba(148,163,184,0.7);
  box-shadow: none;
  color: var(--text);
}

button:hover {
  transform: translateY(-1px);
  filter: brightness(1.03);
}

button:active {
  transform: translateY(0px) scale(0.99);
  box-shadow: 0 8px 20px rgba(37,99,235,0.4);
}

.list-card {
  margin-top: 10px;
  padding: 14px 12px 10px;
  border-radius: var(--radius);
  background: radial-gradient(circle at top, rgba(59,130,246,0.16), transparent 55%),
var(--card);
  border: 1px solid var(--border);
}

.list-header {
  display: flex;
  justify-content: space-between;
  align-items: baseline;
  gap: 8px;
  margin-bottom: 6px;
}

.list-header span {

```

```
    font-size: 12px;
    color: var(--muted);
}

ul {
    list-style: none;
    padding-left: 0;
    margin: 0;
    display: flex;
    flex-direction: column;
    gap: 6px;
    max-height: 260px;
    overflow-y: auto;
}

li {
    padding: 7px 9px;
    border-radius: 9px;
    border: 1px solid rgba(31,41,55,0.95);
    background: rgba(15,23,42,0.92);
    font-size: 14px;
    display: flex;
    justify-content: space-between;
    align-items: center;
    gap: 8px;
}

li span {
    color: var(--muted);
    font-size: 11px;
}

.empty-state {
    font-size: 13px;
    color: var(--muted);
    font-style: italic;
    padding: 4px 0 2px;
}

@media (max-width: 480px) {
    .wrapper {
        padding: 18px 16px 18px;
        border-radius: 18px;
    }
    h1 {
        font-size: 20px;
    }
    .controls {
        flex-direction: column;
    }
    button {
        justify-content: center;
        width: 100%;
    }
}
```



```

    }
  </style>
</head>
<body>
  <div class="wrapper">
    <header>
      <h1>Product List</h1>
      <p>Use the buttons below to add or remove products from the list in real time.</p>

      <div class="controls">
        <button id="addBtn" type="button">Add Item</button>
        <button id="removeBtn" type="button" class="secondary">Remove Item</button>
      </div>
    </header>

    <section class="list-card" aria-label="Product list">
      <div class="list-header">
        <strong>Items</strong>
        <span id="countLabel">3 items</span>
      </div>
      <ul id="productList">
        <li>
          Starter Plan
          <span>#1</span>
        </li>
        <li>
          Growth Plan
          <span>#2</span>
        </li>
        <li>
          Scale Plan
          <span>#3</span>
        </li>
      </ul>
      <div class="empty-state" id="emptyState" style="display:none;">
        No items yet—click “Add Item” to get started.
      </div>
    </section>
  </div>

  <script>
    // Grab references to key DOM elements
    const productList = document.getElementById("productList");
    const countLabel = document.getElementById("countLabel");
    const emptyState = document.getElementById("emptyState");
    const addBtn = document.getElementById("addBtn");
    const removeBtn = document.getElementById("removeBtn");

    // Keep track of how many items we have added so far,
    // so we can label new items with a simple number.
    let itemCounter = productList.children.length;

    // Helper: update the count label and empty-state visibility
    function refreshListMeta() {

```

```

const count = productList.children.length;

// Update the "X items" label
countLabel.textContent = count === 1 ? "1 item" : count + " items";

// Show a friendly empty message when there are no list items
if (count === 0) {
  emptyState.style.display = "block";
} else {
  emptyState.style.display = "none";
}
}

// Add Item handler:
// - Creates a new <li> element
// - Sets its text content
// - Appends it to the <ul>
addBtn.addEventListener("click", function () {
  itemCounter += 1; // Increase our running counter

  // Create a new list item
  const li = document.createElement("li");
  li.textContent = "New Product";

  // Add a small index label to the right
  const meta = document.createElement("span");
  meta.textContent = "#" + itemCounter;
  li.appendChild(meta);

  // Append to the list in the DOM
  productList.appendChild(li);

  // Update count and empty state info
  refreshListMeta();
});

// Remove Item handler:
// - Removes the last <li> from the <ul>, if any exist
removeBtn.addEventListener("click", function () {
  const count = productList.children.length;

  if (count === 0) {
    // Nothing to remove; just return.
    return;
  }

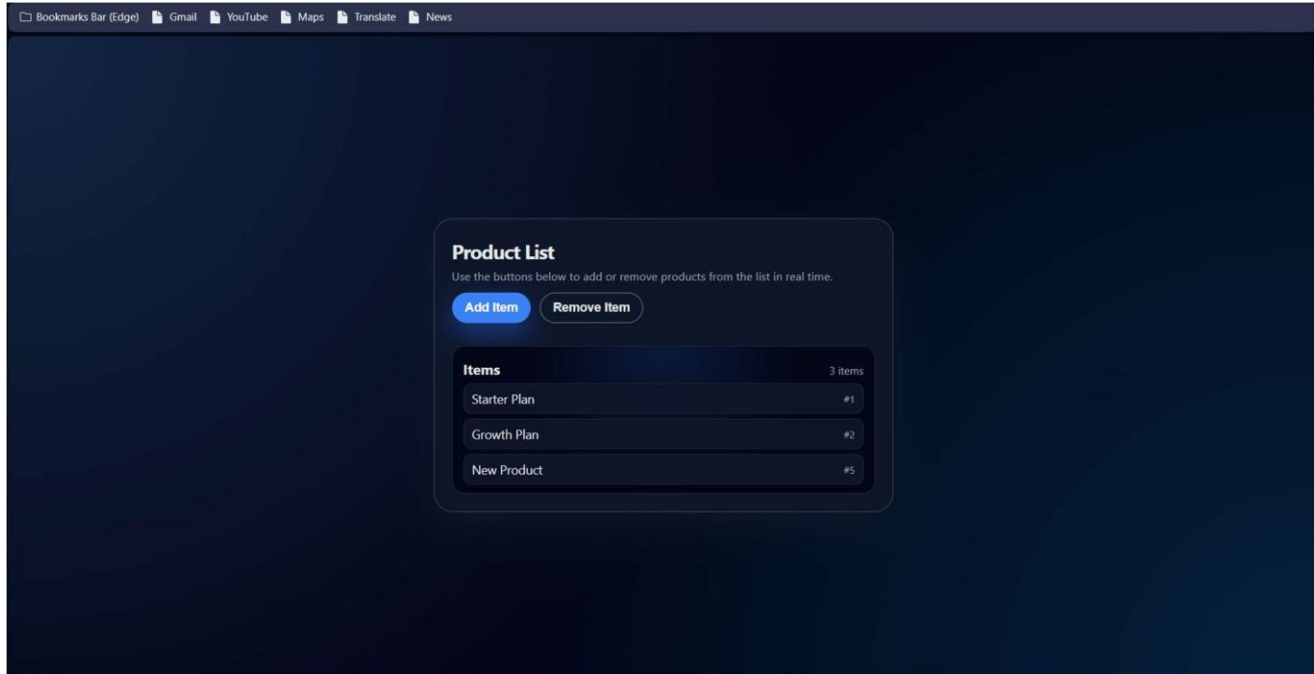
  // Remove the last child (last list item)
  productList.removeChild(productList.lastElementChild);

  // Note: we do not decrement itemCounter here so that
  // newly added items always get a unique number.
  refreshListMeta();
});

```

```
// Initialize meta text on first load
refreshListMeta();
</script>
</body>
</html>
```

Output:



Link:

file:///C:/Users/lilac/OneDrive/Desktop/index4.html

Explanation:

The code uses HTML to create the list structure and buttons, while JavaScript manipulates the DOM to add or remove items dynamically. When a button is clicked, JavaScript updates the list instantly without refreshing the page, making the interaction smooth and interactive.

Task 5: Modal Popup for User Notifications

Prompt:

Create a webpage using HTML, CSS, and JavaScript that includes a modal popup for user notifications or announcements. Design an HTML structure for the modal with a message area and a Close button, and include a separate Open Modal button on the page. Style the modal using CSS with a semi-transparent background overlay so the popup appears centered and clearly above the page content. Use JavaScript to control the modal so that clicking Open Modal displays the popup, and it can be closed either by clicking the Close button or by clicking outside the modal on the overlay. Ensure the popup appears smoothly, the layout is clean and responsive, and the JavaScript code includes clear comments explaining the open and close logic.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
<title>Modal Popup Example</title>
<style>
  /* Basic styling for the page */
  body {
    font-family: Arial, sans-serif;
    margin: 0;
    padding: 20px;
    background-color: #f0f0f0;
  }

  /* Style for the Open Modal button */
  .open-modal-btn {
    padding: 10px 20px;
    font-size: 16px;
    cursor: pointer;
    background-color: #007BFF;
    color: white;
    border: none;
    border-radius: 5px;
  }

  /* Modal overlay styling */
  .modal-overlay {
    position: fixed;
    top: 0;
    left: 0;
    width: 100%;
    height: 100%;
    background-color: rgba(0, 0, 0, 0.5);
    display: none; /* Hidden by default */
    justify-content: center;
    align-items: center;
  }

  /* Modal content styling */
  .modal-content {
    background-color: white;
    padding: 20px;
    border-radius: 5px;
    text-align: center;
    max-width: 400px;
    width: 80%;
  }

  /* Close button styling */
  .close-btn {
    margin-top: 20px;
    padding: 10px 20px;
    font-size: 16px;
    cursor: pointer;
    background-color: #dc3545;
    color: white;
    border: none;
    border-radius: 5px;
  }
</style>
```

```

    }
  </style>
</head>
<body>
  <!-- Open Modal Button -->
  <button class="open-modal-btn">Open Modal</button>
  <!-- Modal Structure -->
  <div class="modal-overlay" id="modalOverlay">
    <div class="modal-content">
      <h2>Notification</h2>
      <p>This is a modal popup for user notifications or announcements.</p>
      <button class="close-btn" id="closeBtn">Close</button>
    </div>
  </div>

  <script>
    // Get the modal overlay and close button elements
    const modalOverlay = document.getElementById('modalOverlay');
    const closeBtn = document.getElementById('closeBtn');
    const openModalBtn = document.querySelector('.open-modal-btn');

    // Function to open the modal
    function openModal() {
      modalOverlay.style.display = 'flex';
    }

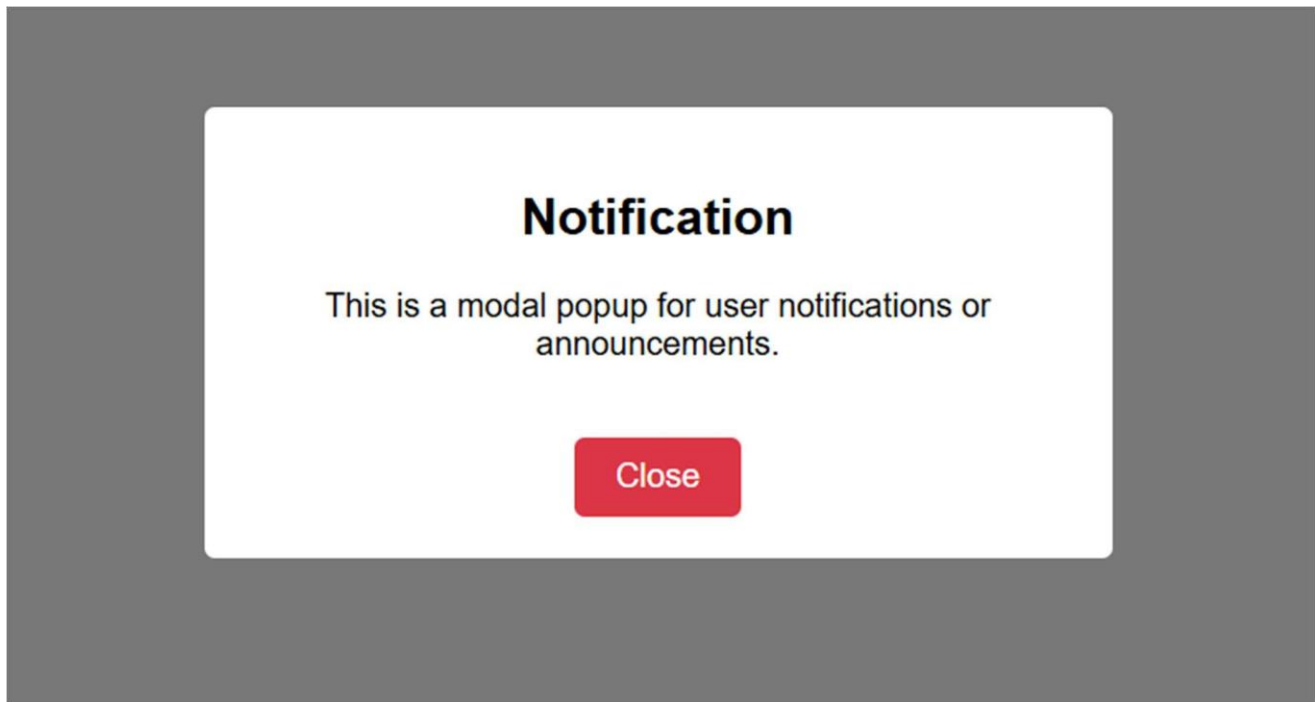
    // Function to close the modal
    function closeModal() {
      modalOverlay.style.display = 'none';
    }

    // Event listener for the Open Modal button
    openModalBtn.addEventListener('click', openModal);

    // Event listener for the Close button
    closeBtn.addEventListener('click', closeModal);

    // Event listener for clicking outside the modal
    modalOverlay.addEventListener('click', function(event) {
      if (event.target === modalOverlay) {
        closeModal();
      }
    });
  </script>
</body>
</html>

```

Output:

Link:

<file:///C:/Users/lilac/OneDrive/Desktop/index5.html>

Explanation:

The code creates a modal popup using HTML for structure and CSS for styling with an overlay effect. JavaScript controls the behavior by opening the modal when the button is clicked and closing it when the user clicks the close button or the overlay. This allows notifications to appear clearly without reloading the page.